

# ABHIJIT SHINDE

## RF & Microwave Engineer — PhD Candidate, Electrical & Computer Engineering

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RF and microwave engineer and ECE PhD candidate with hands-on experience across the full RF development cycle — antenna design, electromagnetic simulation, fabrication, and VNA-based bench validation. Skilled in Ansys HFSS, CST Studio Suite, Keysight ADS, and XFDTD. Seeking a Summer 2027 internship in RF/microwave engineering, antenna design, or wireless communications. Eligible for CPT work authorization (F-1).

## EDUCATION

**Ph.D., Electrical & Computer Engineering**, North Carolina A&T State University | Aug 2023 – Dec 2027 (expected) | GPA 3.68/4.0

**M.E., Electronics & Telecommunication Engineering**, Savitribai Phule Pune University | 2016

**B.E., Electronics & Telecommunication Engineering**, Shivaji University | 2007

*Graduate coursework:* Microwave Circuits · Semiconductor Devices · Estimation Theory · Linear Control Systems · Network Optimization

## TECHNICAL SKILLS

**EM / RF simulation:** Ansys HFSS (Electronics Desktop), CST Studio Suite, Keysight ADS, XFDTD

**RF design:** S-parameters, impedance matching, Smith chart, transmission-line & microstrip modeling; filter design (YIG / tunable / microstrip), couplers, power dividers, matching networks, beamforming

**Antennas:** Vivaldi (UWB), patch, slot, planar arrays; mutual-coupling analysis; SAR / near-field

**Measurement & prototyping:** VNA calibration (SOLT / TRL), return loss / VSWR / gain, anechoic-chamber pattern measurement, PCB prototyping & RF assembly

**Programming:** MATLAB, Python (RF data post-processing, test automation)

## RESEARCH EXPERIENCE

**Ph.D. Researcher — RF, Microwave & Antenna Engineering** North Carolina A&T State University | Aug 2023 – Present

- Designed, fabricated, and bench-tested UWB Vivaldi antenna arrays in Ansys HFSS and XFDTD for wideband (1–7 GHz) microwave imaging; achieved  $|S_{11}| < -10$  dB and  $\sim 4$  dBi realized gain, with reduced mutual coupling through array and feed optimization.
- Developed a planar YIG-based magnetically tunable microwave bandpass filter for frequency-agile RF front ends; presented at IEEE AP-S/URSI 2026.
- Performed S-parameter analysis ( $S_{11}/S_{21}$ ), impedance matching, and SAR / near-field evaluation; correlated measured vs. simulated results to account for fabrication tolerances.
- Built an automated microwave scanning platform (LiteVNA 64, Arduino, MATLAB) for repeatable S-parameter data acquisition.
- Developed CNN-based signal-processing pipelines classifying S-parameter data with >95% accuracy for intracranial hemorrhage detection.

**Graduate coursework projects** North Carolina A&T State University

- Multi-stage broadband impedance-matching network design for microwave circuits; microstrip low-pass filter design with simulation-based validation.
- Dynamic spectrum management and adaptive channel selection for cognitive radio networks.

## TEACHING & INDUSTRY EXPERIENCE

**Assistant Professor, Electrical Engineering** — Sinhgad Institute of Technology (Jul 2022 - May 2023); Dr. D. Y. Patil Institute of Technology (Aug 2017 - Jul 2022). Taught electromagnetics, DSP, image processing, and AI; supervised undergraduate antenna and signal-processing projects.

**Automation Engineer** — KalpTech Solutions (Jan 2014 - Jan 2016); Rapid Pack Engineering (Feb 2008 - Oct 2009). Built ML-based machine-vision inspection systems; programmed industrial control systems.

## PUBLICATIONS & RECOGNITION

- 5 peer-reviewed publications; 10 citations. Presented “Design of a Planar YIG-Tunable Microwave Bandpass Filter” at IEEE AP-S/URSI 2026.
- IEEE Antennas & Propagation Society — Graduate Student Member.
- NASA MUREP reviewer · Best Research Paper Award, ICIC conference · Class of 1969 Endowed Scholarship, NC A&T (2026–27).